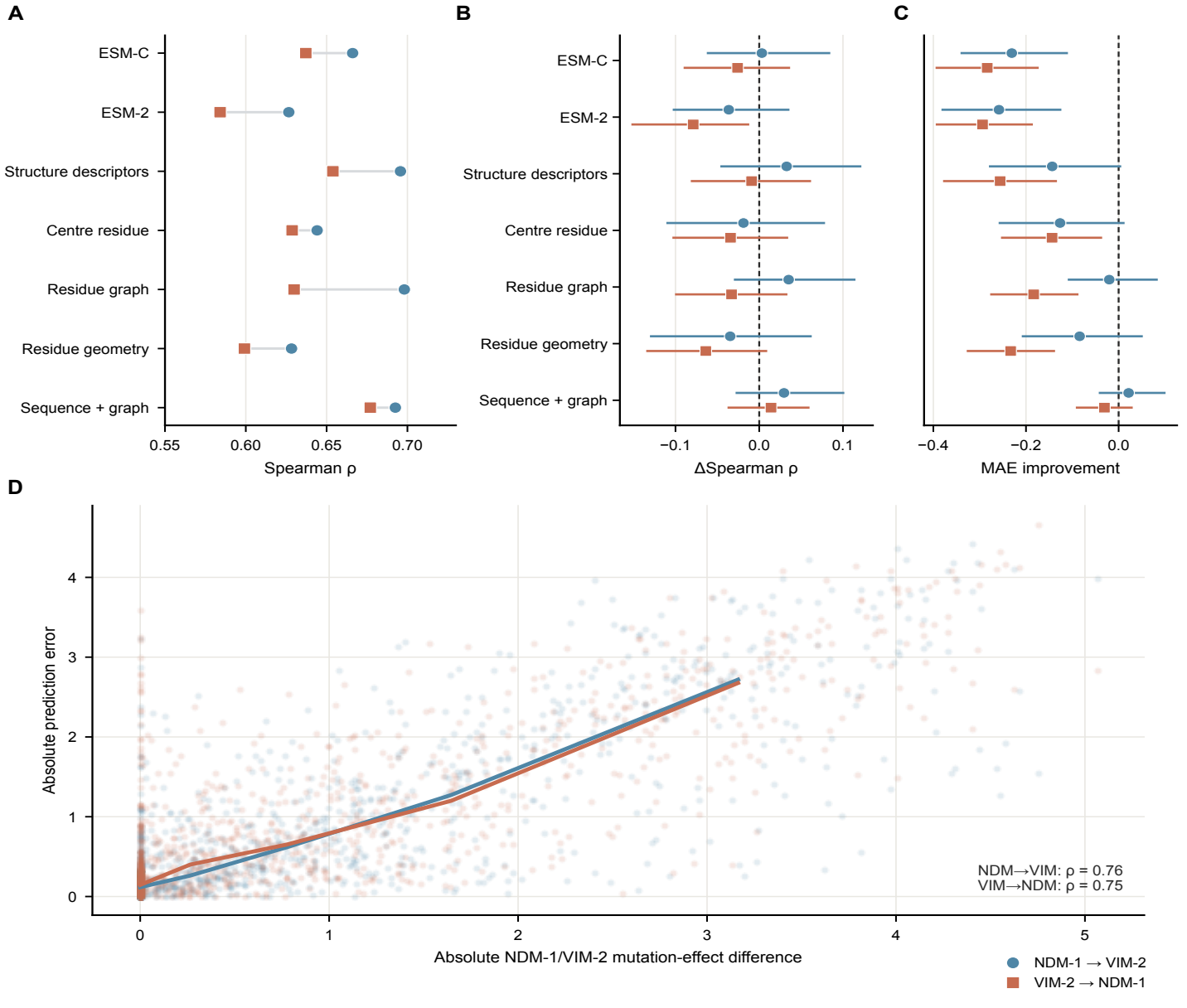

A
B
C
ESM-C
ESM-2
Structure descriptors
Centre residue
Residue graph
Residue geometry
Sequence + graph
0.55
0.60
0.65
0.70
Spearman ρ
−0.1
0.0
0.1
ΔSpearman ρ
−0.4
−0.2
0.0
MAE improvement
D
Absolute prediction error
0
1
2
3
4
5
Absolute NDM-1/VIM-2 mutation-effect difference
NDM→VIM: ρ = 0.76
VIM→NDM: ρ = 0.75
NDM-1 → VIM-2
VIM-2 → NDM-1